JOINTLY DISTRIBUTED RANDOM VARIABLES

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1. Joint & Marginal Distribution Functions
   - Joint & Marginal PMFs of DRVs
   - Joint & Marginal PDFs of CRVs

1. Joint & Marginal Distribution Functions
   - Joint & Marginal PMFs of DRVs
   - Joint & Marginal PDFs of CRVs
2. Independence Among Random Variables
   - Independence Among DRVs
   - Independence Among CRVs

Jointly Distributed Random Variables

- Real life randomnesses possess dependence.
  - X: The odometer value (in km.) of a randomly selected second hand automobile.
  - Y: The second hand price (in ₺) of the same automobile.
  - T: Gender of the last owner of the same automobile ($T = 1$ if female, 0 if male).
- Because X, Y, and T are concerning the same randomly selected automobile, they have a certain dependence.
  - Less the odometer value (X), higher the price (Y).
  - If purchased from female ($T = 1$), higher the price (Y).
  - If purchased from female ($T = 1$), less the odometer value (X).
- To study such dependence among the collection of random variables, we need to use probability statements concerning two or more random variables.

Discrete Case: Joint Probability Mass Functions

- When X and Y are both discrete, the **joint probability mass function** is expressed as:

$$p_{XY}(x,y) = P\{X = x, Y = y\}$$

- Note that, the following equality must always hold:

$$\sum_{(x,y):p_{XY}(x,y)>0} p_{XY}(x,y) = 1$$

Discrete Case - Marginal Probability Mass Functions

- The marginal PMF of X can be obtained by:

$$\begin{aligned} p_X(x) &= P\{X = x\} \\ &= P\{X = x, -\infty < Y < \infty\} \\ &= \sum_{y:p_{XY}(x,y)>0} P\{X = x, Y = y\} \\ &= \sum_{y:p_{XY}(x,y)>0} p_{XY}(x,y) \end{aligned}$$

- The marginal PMF of Y can be obtained by:

$$\begin{aligned} p_Y(y) &= P\{Y = y\} \\ &= P\{-\infty < X < \infty, Y = y\} \\ &= \sum_{x:p_{XY}(x,y)>0} P\{X = x, Y = y\} \\ &= \sum_{x:p_{XY}(x,y)>0} p_{XY}(x,y) \end{aligned}$$

Discrete Case - Joint & Marginal PMFs: Example

Q1. Suppose that X and Y have the following joint PMF:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. Compute the marginal PMF $p_X(x)$ of X.

b. Compute the marginal PMF $p_Y(y)$ of Y.

c. Compute $P\{X=3\}$ and $P\{Y=2\}$.

d. Compute $E\left[(X-2)^2\right]$.

Discrete Case - Joint & Marginal PMFs: Example (cont'd)

A1. We will use the table to answer the questions:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

a. Row sums will yield $p_X(x)$:

$$p_X(x) = \sum_{y\in\{0,2,4\}} p_{XY}(x,y) = \begin{cases} 0.20+0.20+0.10=0.50, & x=1 \\ 0.10+0.05+0.10=0.25, & x=2 \\ 0.10+0.05+0.10=0.25, & x=3 \\ 0, & \text{e/w} \end{cases}$$

Discrete Case - Joint & Marginal PMFs: Example (cont'd)

A1. We will use the table to answer the questions:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

b. Column sums will yield $p_Y(y)$:

$$p_Y(y) = \sum_{x=1}^{3} p_{XY}(x,y) = \begin{cases} 0.20+0.10+0.10 = 0.40, & y=0 \\ 0.20+0.05+0.05 = 0.30, & y=2 \\ 0.10+0.10+0.10 = 0.30, & y=4 \\ 0, & \text{e/w} \end{cases}$$

Discrete Case - Joint & Marginal PMFs: Example (cont'd)

A1. We will use the table to answer the questions:

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

c. Marginal PMFs tell us that:

$$P\{X=3\}=0.25, \qquad P\{Y=2\}=0.3$$

d. The requested expectation is:

$$\begin{aligned} E\left[(X-2)^2\right] &= \sum_{x=1}^{3}(x-2)^2 p_X(x) \\ &= 1\times 0.5+0\times 0.25+1\times 0.25=0.75 \end{aligned}$$

Discrete Case - Joint & Marginal PMFs: Example II

Q2. Suppose that 2 balls are randomly selected from an urn containing 2 red, 3 green, and 4 blue balls. If we let X and Y denote, respectively, the number of red and green balls chosen:

a. Write down and tabulate the joint PMF of X and Y.
b. Write down the marginal PMF of X.
c. Write down the marginal PMF of Y.
d. Calculate the expected number of red balls selected.
e. Calculate the expected number of green balls selected.

A2. All probabilities in this question can be computed using multinomial coefficients[1]. For example:

$$p_{XY}(1,1) = P\{X=1, Y=1\} = \frac{\binom{2}{1}\binom{3}{1}\binom{4}{1}}{\binom{9}{2}} = \frac{6}{36}$$

[1] See page 39 of "LS01.pdf" file.

Discrete Case - Joint & Marginal PMFs: Example II (cont'd)

a. The joint PMF can be written and tabulated as follows:

$$p_{XY}(x,y) = \begin{cases} 6/36, & x=0, y=0 \\ 12/36, & x=0, y=1 \\ 3/36, & x=0, y=2 \\ 8/36, & x=1, y=0 \\ 6/36, & x=1, y=1 \\ 1/36, & x=2, y=0 \\ 0, & \text{elsewhere} \end{cases}$$

$p_{XY}(x,y)$	y		
x	0	1	2
0	$\frac{6}{36}$	$\frac{12}{36}$	$\frac{3}{36}$
1	$\frac{8}{36}$	$\frac{6}{36}$	0
2	$\frac{1}{36}$	0	0

Discrete Case - Joint & Marginal PMFs: Example II (cont'd)

b. The marginal PMF of X can be obtained through column sums of the tabulated joint PMF.

$p_{XY}(x,y)$	y		
x	0	1	2
0	$\frac{6}{36}$	$\frac{12}{36}$	$\frac{3}{36}$
1	$\frac{8}{36}$	$\frac{6}{36}$	0
2	$\frac{1}{36}$	0	0

$$p_X(x) = \sum_{y=0}^{2} p_{XY}(x,y) = \begin{cases} \frac{6}{36} + \frac{12}{36} + \frac{3}{36} = \frac{21}{36}, & x = 0 \\ \frac{8}{36} + \frac{6}{36} = \frac{14}{36}, & x = 1 \\ \frac{1}{36}, & x = 2 \\ 0, & \text{e/w} \end{cases}$$

Discrete Case - Joint & Marginal PMFs: Example II (cont'd)

c. The marginal PMF of Y can be obtained through column sums of the tabulated joint PMF.

$p_{XY}(x,y)$	y		
x	0	1	2
0	$\frac{6}{36}$	$\frac{12}{36}$	$\frac{3}{36}$
1	$\frac{8}{36}$	$\frac{6}{36}$	0
2	$\frac{1}{36}$	0	0

$$p_Y(y) = \sum_{x=0}^{2} p_{XY}(x,y) = \begin{cases} \frac{6}{36} + \frac{8}{36} + \frac{1}{36} = \frac{15}{36}, & y = 0 \\ \frac{12}{36} + \frac{6}{36} = \frac{18}{36}, & y = 1 \\ \frac{3}{36}, & y = 2 \\ 0, & \text{e/w} \end{cases}$$

Discrete Case - Joint & Marginal PMFs: Example II (cont'd)

d. The marginal PMF of X can be helpful to find the expected number of red balls selected, namely $E[X]$.

$$p_X(x) = \begin{cases} \frac{21}{36}, & x = 0 \\ \frac{14}{36}, & x = 1 \\ \frac{1}{36}, & x = 2 \\ 0, & \text{e/w} \end{cases}$$

$$E[X] = 0 \times \frac{21}{36} + 1 \times \frac{14}{36} + 2 \times \frac{1}{36} = \frac{16}{36} \approx 0.444 \,\text{red balls}$$

Discrete Case - Joint & Marginal PMFs: Example II (cont'd)

e. The marginal PMF of Y can be helpful to find the expected number of green balls selected, namely $E[Y]$.

$$p_Y(y) = \begin{cases} \frac{15}{36}, & y = 0 \\ \frac{18}{36}, & y = 1 \\ \frac{3}{36}, & y = 2 \\ 0, & \text{e/w} \end{cases}$$

$$E[Y] = 0 \times \frac{15}{36} + 1 \times \frac{18}{36} + 2 \times \frac{3}{36} = \frac{24}{36} \approx 0.666 \,\text{green balls}$$

Continuous Case - Joint Probability Density Functions

- X and Y are jointly continuous RVs if there exists a non-negative **joint probability density function** $f_{XY}(x,y)$, defined for all real x and y, having the property that, for any couple of real values a and b:

$$P\{X \leq a, Y \leq b\} = F_{XY}(a,b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f_{XY}(x,y)\,dxdy$$

- Note that, the following equation must hold:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y)\,dxdy = 1$$

  The volume under the joint PDF should be equal to 1.

- It follows upon differentiation that:

$$\frac{\partial^2 F_{XY}(x,y)}{\partial x \partial y} = f_{XY}(x,y)$$

Continuous Case - Marginal Probability Density Functions

- If X and Y are jointly continuous, they are also marginally continuous.
- The **marginal probability density functions** can be obtained similarly as in their discrete counterpart.
- The marginal PDF of X:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$$

- The marginal PDF of Y:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dx$$

Cont. Case - Marginal Cumulative Distribution Functions

- Once the marginal PDF of X is computed, its marginal CDF F_X can also be computed using the traditional approach.

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

- The marginal CDF of Y can be computed in a similar fashion.

Continuous Case - Joint & Marginal PDFs: Example

Q3. For a randomly selected freshman student, let X and Y be the fractions of class participation in PHYS 181 and PHYS 184 classes, respectively. The joint probability density function of X and Y is:

$$f_{XY}(x,y) = \begin{cases} cxy, & 0 < x < 1,\ 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Determine c so that f_{XY} is a joint PDF.
b. Find $P\{0 \leq X \leq 0.5,\ 0.25 \leq Y \leq 0.5\}$.
c. Find the marginal PDF, $f_X(x)$ of X and compute $P\{X \leq 0.7\}$.
d. Find the marginal PDF, $f_Y(y)$ of Y and compute $P\{0.1 \leq Y \leq 0.4\}$.

A3. First, we need to solve (a) so we can solve (b), (c), and (d).

Continuous Case - Joint & Marginal PDFs: Ex. (cont'd)

a. In order to have a joint PDF, the volume under the f_{XY} must be equal to 1.

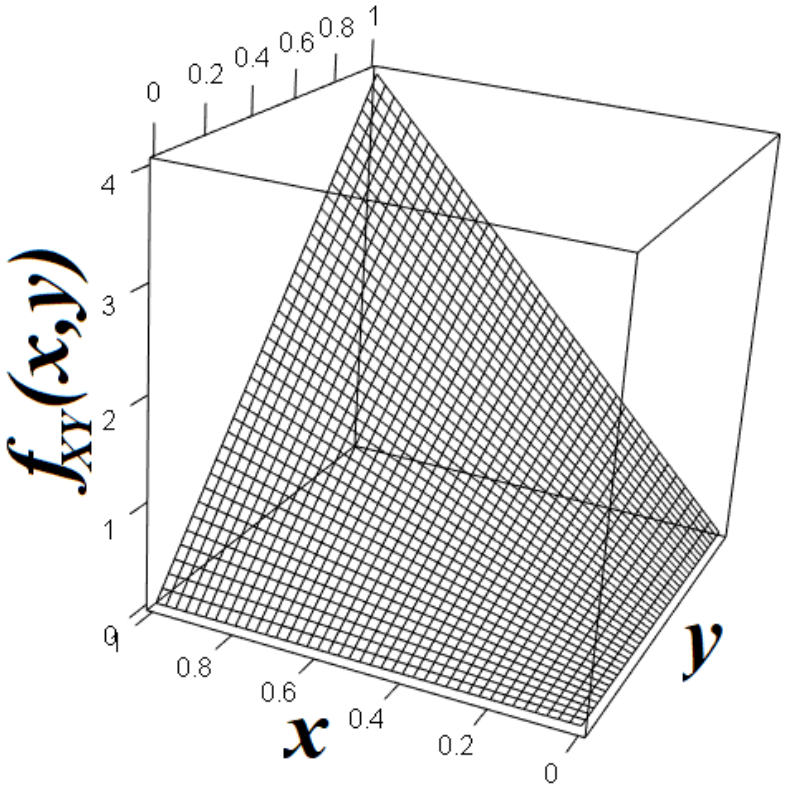
$$\begin{aligned}
\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)\,dxdy &= \int_0^1\int_0^1 cxydxdy \\
&= \int_0^1 \left[c\frac{x^2}{2}y\right]_{x=0}^{x=1} dy \\
&= \int_0^1 \frac{c}{2}ydy \\
&= \left.\frac{cy^2}{4}\right|_{y=0}^{y=1} \\
&= \frac{c}{4} = 1
\end{aligned}$$

Therefore, we must have $c = 4$.

Continuous Case - Joint & Marginal PDFs: Ex. (cont'd)

a. Let us illustrate the joint PDF using a 3D plot diagram.

$$f_{XY}(x,y) = \begin{cases} 4xy, & 0 < x < 1,\ 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$



Continuous Case - Joint & Marginal PDFs: Ex. (cont'd)

b. The requested probability is:

$$
\begin{aligned}
P\{0 \leq X \leq 0.5, 0.25 \leq Y \leq 0.5\} &= \int_{0.25}^{0.5} \int_{0}^{0.5} f_{XY}(x,y)\,dxdy \\
&= \int_{0.25}^{0.5} \int_{0}^{0.5} 4xydxdy \\
&= \int_{0.25}^{0.5} \left[2x^2y\right]_{x=0}^{x=0.5} dy \\
&= \int_{0.25}^{0.5} \frac{y}{2} dy \\
&= \left[\frac{y^2}{4}\right]_{y=0.25}^{y=0.5} \\
&= \left[\frac{0.25}{4}\right] - \left[\frac{0.0625}{4}\right] \approx 0.0469
\end{aligned}
$$

Continuous Case - Joint & Marginal PDFs: Ex. (cont'd)

c. The marginal PDF of X:

$$f_X(x) = \int_0^1 f_{XY}(x,y)\,dy = \int_0^1 4xy\,dy = \left[2xy^2\right]_{y=0}^{y=1} = 2x, \quad x \in [0,1]$$

The requested probability is:

$$P\{X \leq 0.7\} = \int_0^{0.7} f_X(x)\,dx = \int_0^{0.7} 2x\,dx = \left[x^2\right]_{x=0}^{x=0.7} = 0.49$$

Continuous Case - Joint & Marginal PDFs: Ex. (cont'd)

d. The marginal PDF of Y:

$$f_Y(y) = \int_0^1 f_{XY}(x,y)\,dx = \int_0^1 4xy\,dx = \left[2x^2y\right]_{x=0}^{x=1} = 2y, \quad y \in [0,1]$$

The requested probability is:

$$P\{0.1 \le Y \le 0.4\} = \int_{0.1}^{0.4} f_Y(y)\,dy = \int_{0.1}^{0.4} 2y\,dy = \left[y^2\right]_{y=0.1}^{y=0.4} = 0.15$$

Continuous Case - Joint & Marginal PDFs: Example II

Q4. Consider the following joint PDF of X and Y:

$$f_{XY}(x,y) = 12x^2y^3, \quad 0 < x < 1,\ 0 < y < 1$$

Compute the probability $P\{X \leq Y\}$.

A4. Note that, according to the probability statement:

- X can take any value in $(0,1)$ since Y is an upper bound for X and $Y \in (0,1)$.
- However, for fixed $X = x$, Y can only take values in $[x,1)$.

Thus, the probability can be computed as:

$$P\{X \leq Y\} = \int_0^1 \int_x^1 12x^2y^3 dydx$$

- The operator of the outer integral is x, taking any value in $(0,1)$.
- The operator of the inner integral is y, taking any value in $[x,1)$.

Continuous Case - Joint & Marginal PDFs: Ex. II (cont'd)

A4. The solution is:

$$\begin{aligned} P\{X \leq Y\} &= \int_0^1 \int_{y=x}^{y=1} 12x^2y^3 dydx = \int_0^1 \left[3x^2y^4\right]_x^1 dx \\ &= \int_0^1 \left[3x^2 - 3x^6\right] dx = \left[x^3 - \tfrac{3}{7}x^7\right]_{x=0}^{x=1} = \tfrac{4}{7} \end{aligned}$$

or we can interchange the integral operators to get the same result. But do not forget that, this time:

- The operator of the outer integral is y, taking any value in $(0,1)$.
- The operator of the inner integral is x, taking any value in $(0,y]$.

$$\begin{aligned} P\{X \leq Y\} &= \int_0^1 \int_0^y 12x^2y^3 dxdy = \int_0^1 \left[4x^3y^3\right]_{x=0}^{x=y} dy \\ &= \int_0^1 4y^6 dy = \left[\tfrac{4}{7}y^7\right]_{y=0}^{y=1} = \tfrac{4}{7} \end{aligned}$$

Continuous Case - Joint & Marginal PDFs: Example III

Q5. Let X and Y denote the profitability factor of two different investments. The joint PDF of these RVs is:

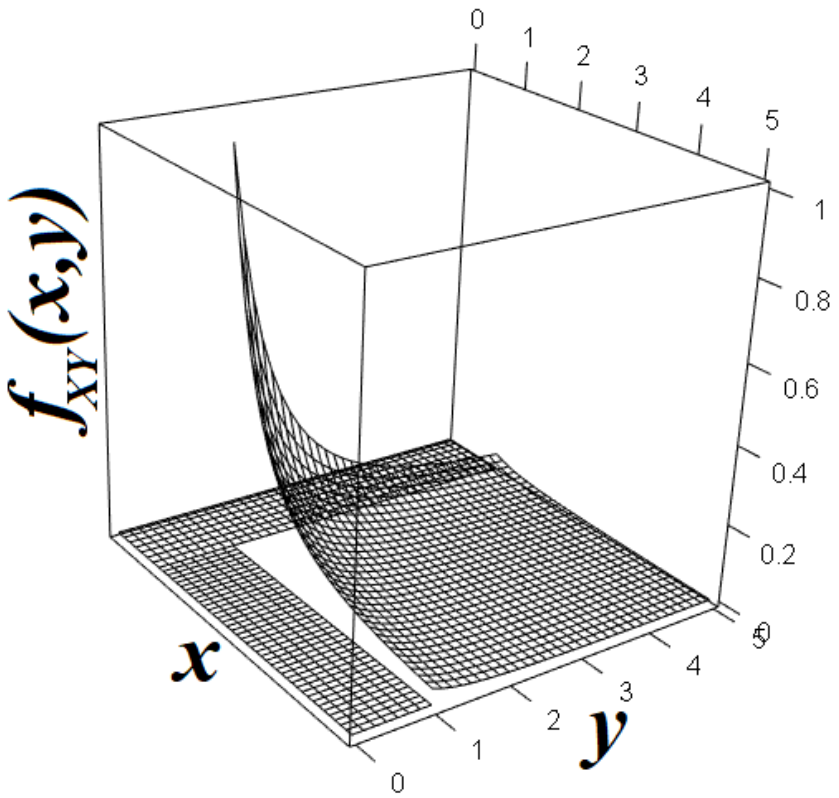
$$f_{XY}(x,y) = \begin{cases} \frac{1}{x^2y^2}, & x \geq 1,\ y \geq 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Find the marginal PDF and marginal CDF of X and compute the probability $P\{1.2 \leq X \leq 1.5\}$.
b. Find the marginal PDF and marginal CDF of Y and compute the probability $P\{2 \leq Y \leq 2.5\}$.
c. Find probability $P\{X \geq 2Y\}$.

A5. Before solving the question, let us illustrate the joint PDF using a 3D plot diagram.

Continuous Case - Joint & Marginal PDFs: Ex. III (cont'd)

$$f_{XY}(x,y) = \begin{cases} \frac{1}{x^2y^2}, & x \geq 1,\ y \geq 1 \\ 0, & \text{elsewhere} \end{cases}$$



Continuous Case - Joint & Marginal PDFs: Ex. III (cont'd)

a. The marginal PDF of X is:

$$\begin{aligned} f_X(x) &= \int_1^\infty f_{XY}(x,y)\,dy = \int_1^\infty x^{-2}y^{-2}dy = \left[-x^{-2}y^{-1}\right]_{y=1}^{y=\infty} \\ &= \left[-\frac{1}{x^2y}\right]_{y=1}^{y=\infty} = [0] - \left[-\frac{1}{x^2}\right] = \frac{1}{x^2}, \quad x \geq 1 \end{aligned}$$

The marginal CDF of X is:

$$F_X(x) = \int_1^x \frac{1}{t^2}dt = \left[-\frac{1}{t}\right]_1^x = 1 - \frac{1}{x}, \quad x \geq 1$$

$$\begin{aligned} P\{1.2 \leq X \leq 1.5\} &= F_X(1.5) - F_X(1.2) \\ &= \left(1 - \frac{1}{1.5}\right) - \left(1 - \frac{1}{1.2}\right) = \frac{1}{6} \end{aligned}$$

Continuous Case - Joint & Marginal PDFs: Ex. III (cont'd)

b. Notice that the marginal PDF and CDF of Y should be the same of X, since their influence in the joint PDF is the same:

$$f_{XY}(x,y) = \begin{cases} \frac{1}{x^2y^2}, & x \geq 1,\ y \geq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Hence:

$$f_Y(y) = \frac{1}{y^2}, \quad F_Y(y) = 1 - \frac{1}{y}, \quad y \geq 1$$

The probability of interest is:

$$\begin{aligned} P\{2 \leq Y \leq 2.5\} &= F_Y(2.5) - F_Y(2) \\ &= \left(1 - \frac{1}{2.5}\right) - \left(1 - \frac{1}{2}\right) = \frac{1}{10} \end{aligned}$$

Continuous Case - Joint & Marginal PDFs: Ex. III (cont'd)

c. In order to find the probability of an inequality statement involving two random variables we need to use the joint PDF.

$$P\{X \geq 2Y\} = \int_{2}^{\infty} \int_{1}^{x/2} \frac{1}{x^2 y^2} dy dx$$

- The operator of the outer integral is x which can take values greater than 2 according to the prob. statement.
- The operator of the inner integral is y which can take values between 1 and $x/2$ according to the prob. statement.

$$\begin{aligned} P\{X \geq 2Y\} &= \int_{2}^{\infty} \int_{1}^{x/2} x^{-2} y^{-2} dy dx = \int_{2}^{\infty} \left[-x^{-2} y^{-1}\right]_{y=1}^{y=x/2} dx \\ &= \int_{2}^{\infty} \left[\left(-2x^{-3}\right) - \left(-x^{-2}\right)\right] dx = \int_{2}^{\infty} x^{-2} - 2x^{-3} dx \\ &= \left[-x^{-1} + x^{-2}\right]_{x=2}^{\infty} = 0 - \left(-\tfrac{1}{2} + \tfrac{1}{4}\right) = \tfrac{1}{4} \end{aligned}$$

Continuous Case - Joint & Marginal PDFs: Example IV

Q6. Let X and Y be two continuous RVs with the joint PDF:

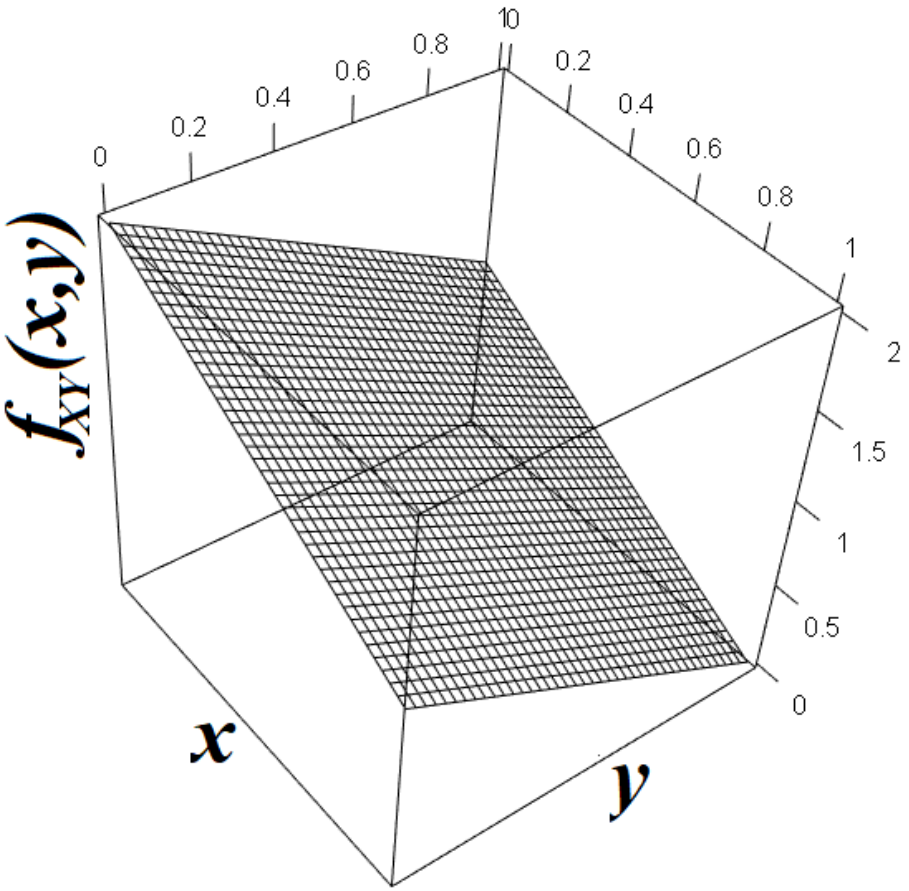
$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Find the marginal PDF and marginal CDF of X and compute the probability $P\{X \leq 0.5\}$.
b. Compute $E[X]$.
c. Compute $Var[X]$.

A6. Before solving the question, let us illustrate the joint PDF using a 3D plot diagram.

Continuous Case - Joint & Marginal PDFs: Ex. IV (cont'd)

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{elsewhere} \end{cases}$$



Continuous Case - Joint & Marginal PDFs: Ex. IV (cont'd)

a. The marginal PDF of X is:

$$\begin{aligned} f_X(x) &= \int_0^1 f_{XY}(x,y)\,dy = \int_0^1 (2-x-y)\,dy \\ &= \left[2y - xy - \frac{y^2}{2}\right]_{y=0}^{y=1} = \frac{3}{2} - x, \quad x \in [0,1] \end{aligned}$$

The marginal CDF is:

$$F_X(x) = \int_0^x \left(\frac{3}{2} - t\right) dt = \left[\frac{3t}{2} - \frac{t^2}{2}\right]_0^x = \frac{3x - x^2}{2}, \quad x \in [0,1]$$

The requested probability is:

$$P\{X \leq 0.5\} = F_X(0.5) = \frac{(3)(0.5) - (0.5)^2}{2} = \frac{5}{8}$$

Note that the marginal PDF and the marginal CDF of Y should also be the same of X.

Continuous Case - Joint & Marginal PDFs: Ex. IV (cont'd)

b. The expected value of X can be computed once its marginal PDF is found.

$$\begin{aligned} E[X] &= \int_0^1 x f_X(x)\,dx = \int_0^1 \left(\tfrac{3x}{2} - x^2\right) dx \\ &= \left[\tfrac{3x^2}{4} - \tfrac{x^3}{3}\right]_{x=0}^{x=1} = \tfrac{3}{4} - \tfrac{1}{3} = \tfrac{5}{12} \end{aligned}$$

c. The variance of X can be computed as follows:

$$\begin{aligned} E\left[X^2\right] &= \int_0^1 x^2 f_X(x)\,dx = \int_0^1 \left(\tfrac{3x^2}{2} - x^3\right) dx \\ &= \left[\tfrac{x^3}{2} - \tfrac{x^4}{4}\right]_{x=0}^{x=1} = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4} \end{aligned}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \tfrac{1}{4} - \left(\tfrac{5}{12}\right)^2 = \tfrac{11}{144}$$

Continuous Case - Joint & Marginal PDFs: Example V

Q7. Let X and Y be two continuous RVs with the joint PDF:

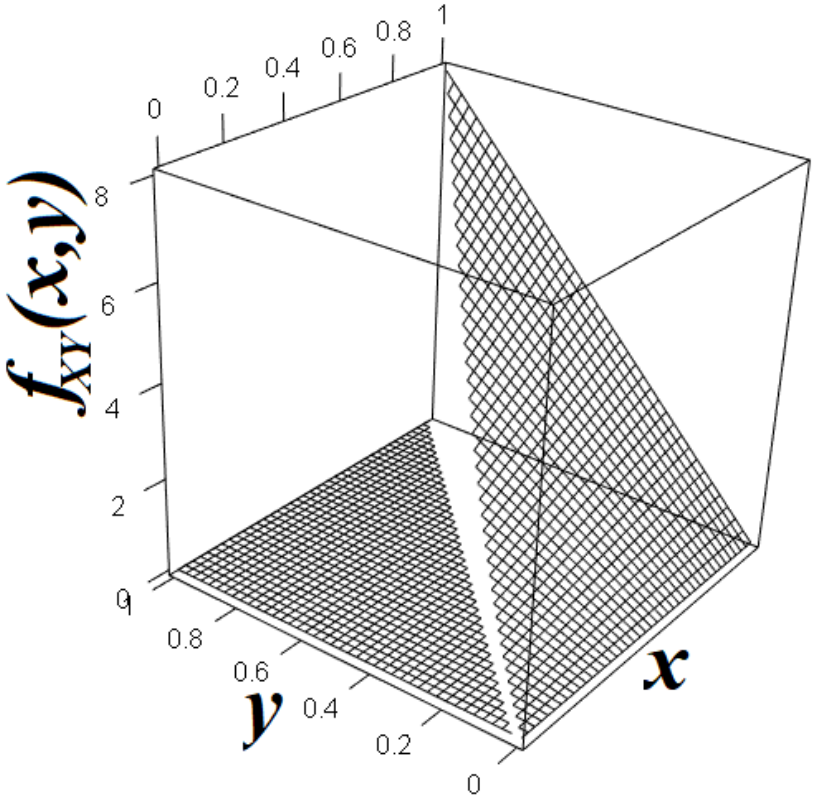
$$f_{XY}(x,y) = \begin{cases} 8xy, & 0 \leq x \leq 1,\ 0 \leq y \leq x \\ 0, & \text{elsewhere} \end{cases}$$

a. Compute the marginal PDF X.
b. Compute the marginal PDF Y.

A7. Notice that, Y is bounded by X. In other words, Y can never be greater than X. However, since X can take any value between 0 and 1, Y can take values up to 1, as long as X is large enough. Let us illustrate the joint PDF using a 3D plot diagram.

Continuous Case - Joint & Marginal PDFs: Ex. V (cont'd)

$$f_{XY}(x,y) = \begin{cases} 8xy, & 0 \le x \le 1,\ 0 \le y \le x \\ 0, & \text{elsewhere} \end{cases}$$



Continuous Case - Joint & Marginal PDFs: Ex. V (cont'd)

a. Let us compute the marginal PDF of X. While integrating the joint PDF over y, notice that $0 \leq y \leq x$.

$$f_X(x) = \int_0^x 8xydy = \left[4xy^2\right]_{y=0}^{y=x} = 4x^3, \quad x \in [0,1]$$

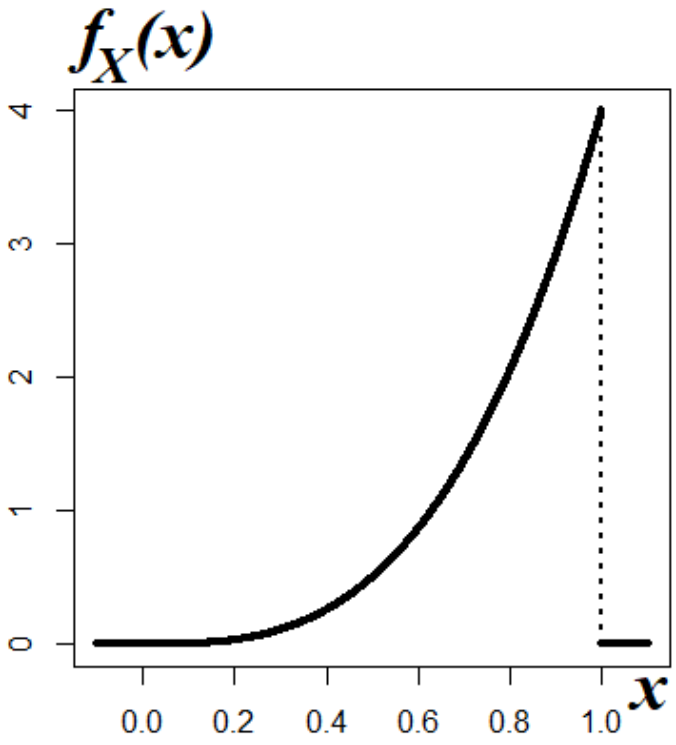
b. Let us compute the marginal PDF of Y. While integrating the joint PDF over x, notice that $y \leq x \leq 1$.

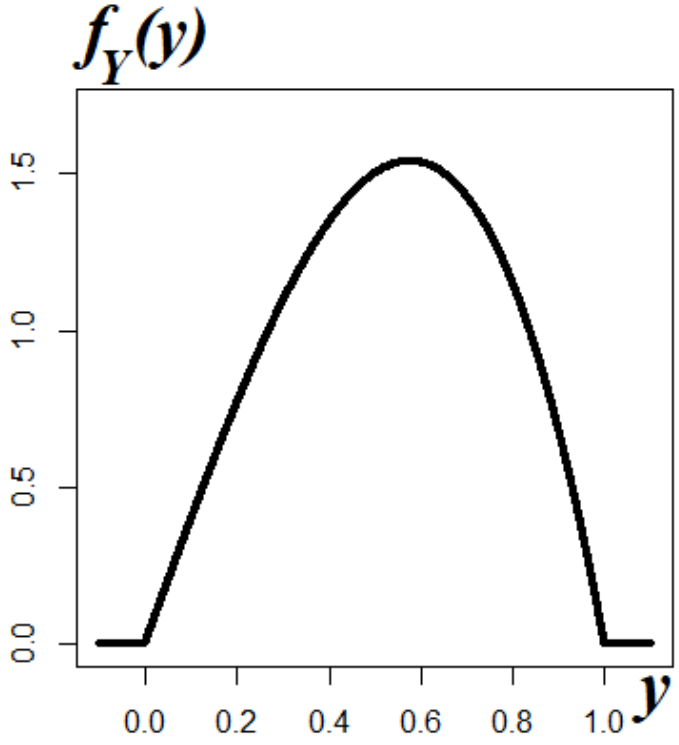
$$f_Y(y) = \int_y^1 8xydx = \left[4x^2y\right]_{x=y}^{x=1} = 4y - 4y^3, \quad y \in [0,1]$$

Let us illustrate $f_X(x)$ and $f_Y(y)$ in a scatter plot.

Continuous Case - Joint & Marginal PDFs: Ex. V (cont'd)

$$f_X(x) = 4x^3, \quad x \in [0,1], \qquad f_Y(y) = 4y - 4y^3, \quad y \in [0,1]$$





Independence Among Random Variables

- The random variables X and Y are said to be independent if and only if, for all real numbers a and b:

$$P\{X \leq a, Y \leq b\} = P\{X \leq a\} P\{Y \leq b\}$$

- Note that, the above expression must hold for all a and b real values, so that we can claim X and Y to be independent.

Independence Among RVs - Discrete Case

- When X and Y are **discrete** random variables, the condition of independence in the above equation is equivalent to:

$$p_{XY}(x,y) = p_X(x)\, p_Y(y)$$

  for all real values of x and y.
- If, for any real couple of x and y:

$$p_{XY}(x,y) \neq p_X(x)\, p_Y(y)$$

  then, we say X and Y are not independent.

Independence Among RVs - Discrete Case: Example

Q8. Let X and Y be two Bernoulli random variables with joint probability mass function.

$$p_{XY}(x,y) = \begin{cases} 0.18, & x=0,\, y=0 \\ 0.12, & x=0,\, y=1 \\ 4k, & x=1,\, y=0 \\ 3k, & x=1,\, y=1 \\ 0, & \text{elsewhere} \end{cases}$$

Are X and Y independent?

Independence Among RVs - Discrete Case: Example

Q8. Let X and Y be two Bernoulli random variables with joint probability mass function.

$$p_{XY}(x,y) = \begin{cases} 0.18, & x=0,\, y=0 \\ 0.12, & x=0,\, y=1 \\ 4k, & x=1,\, y=0 \\ 3k, & x=1,\, y=1 \\ 0, & \text{elsewhere} \end{cases}$$

Are X and Y independent?

A8. In order to answer this question, first we need to find k. Since both X and Y are discrete, the probabilities for all possible values should sum up to 1. This means

$$0.18 + 0.12 + 4k + 3k = 1$$

which indicates $k = 0.1$. Now we need to find the marginal PMFs of X and Y.

Independence Among RVs - Discrete Case: Ex. (cont'd)

A8. The joint PMF is given as follows in the tabulated format:

$p_{XY}(x,y)$	y	
x	0	1
0	0.18	0.12
1	0.40	0.30

The marginal distributions, then, are as follows:

$$p_X(x) = \begin{cases} 0.3, & x = 0 \\ 0.7, & x = 1 \\ 0, & \text{e/w} \end{cases}, \quad p_Y(y) = \begin{cases} 0.58, & y = 0 \\ 0.42, & y = 1 \\ 0, & \text{e/w} \end{cases}$$

Independence Among RVs - Discrete Case: Ex. (cont'd)

A8. Obviously, $p_{XY}(x,y) = p_X(x)\,p_Y(y)$ does not hold for all possible values for example:

$$p_X(0)\,p_Y(0) = 0.3 \times 0.58 = 0.174 \neq p_{XY}(0,0) = 0.18$$

Since the independence equation does not hold for this instance, we say that X and Y are **not independent**.

Independence Among RVs - Discrete Case: Example II

Q9. Consider the joint PMF of two Bernoulli RVs, X and Y, given in the tabulated format:

$p_{XY}(x,y)$	y	
x	0	1
0	0.10	0.30
1	0.15	0.45

Are X and Y independent:

Independence Among RVs - Discrete Case: Example II

Q9. Consider the joint PMF of two Bernoulli RVs, X and Y, given in the tabulated format:

$p_{XY}(x,y)$	y	
x	0	1
0	0.10	0.30
1	0.15	0.45

Are X and Y independent:

A9. The marginal distributions, then, are as follows:

$$p_X(x) = \begin{cases} 0.4, & x = 0 \\ 0.6, & x = 1 \\ 0, & \text{e/w} \end{cases}, \quad p_Y(y) = \begin{cases} 0.25, & y = 0 \\ 0.75, & y = 1 \\ 0, & \text{e/w} \end{cases}$$

Independence Among RVs - Discrete Case: Ex. II (cont'd)

A9. The independence condition (for DRVs) is:

$$p_{XY}(x,y) = p_X(x)\, p_Y(y), \quad \forall x,y$$

Let us check if this condition holds:

$$p_X(0)\, p_Y(0) = 0.4 \times 0.25 = 0.10 = p_{XY}(0,0) = 0.10$$

$$p_X(0)\, p_Y(1) = 0.4 \times 0.75 = 0.30 = p_{XY}(0,1) = 0.30$$

$$p_X(1)\, p_Y(0) = 0.6 \times 0.25 = 0.15 = p_{XY}(1,0) = 0.15$$

$$p_X(1)\, p_Y(1) = 0.6 \times 0.75 = 0.45 = p_{XY}(1,1) = 0.45$$

Since the independence equation holds for all instances, we say that X and Y are **independent**.

Independence Among RVs - Discrete Case: Example III

Q10. Consider the joint PMF of X and Y provided in Question 1.

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

Are X and Y independent:

Independence Among RVs - Discrete Case: Example III

Q10. Consider the joint PMF of X and Y provided in Question 1.

$p_{XY}(x,y)$	y		
x	0	2	4
1	0.20	0.20	0.10
2	0.10	0.05	0.10
3	0.10	0.05	0.10

Are X and Y independent:

A10. Observe that:

$$p_X(1)\,p_Y(0) = 0.50 \times 0.40 = 0.2 = p_{XY}(1,0) = 0.20$$

However:

$$p_X(1)\,p_Y(2) = 0.50 \times 0.30 = 0.15 \neq p_{XY}(1,2) = 0.20$$

Therefore, X and Y are not independent.

Independence Among RVs: Continuous Case

- When X and Y are **continuous** random variables, the condition of independence in the above equation is equivalent to:

$$f_{XY}(x,y) = f_X(x) f_Y(y)$$

for all real values of x and y.

- If, for any real couple of x and y:

$$f_{XY}(x,y) \neq f_X(x) f_Y(y)$$

then, we say X and Y are not independent.

Independence Among RVs - Continuous Case: Example

Q11. Consider the CRVs X and Y in Question 6 with the joint PDF:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

for which we have computed the marginal PDFs as:

$$f_X(x) = \frac{3}{2} - x, \quad x \in [0,1]$$
$$f_Y(y) = \frac{3}{2} - y, \quad y \in [0,1]$$

Are X and Y independent?

Independence Among RVs - Continuous Case: Example

Q11. Consider the CRVs X and Y in Question 6 with the joint PDF:

$$f_{XY}(x,y) = \begin{cases} 2-x-y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{elsewhere} \end{cases}$$

for which we have computed the marginal PDFs as:

$$f_X(x) = \frac{3}{2} - x, \quad x \in [0,1]$$
$$f_Y(y) = \frac{3}{2} - y, \quad y \in [0,1]$$

Are X and Y independent?

A11. We need to check the independence condition.

$$f_X(x) f_Y(y) = \left(\frac{3}{2} - x\right)\left(\frac{3}{2} - y\right) = \frac{9}{4} - \frac{3}{2}x - \frac{3}{2}y + xy$$
$$f_{XY}(x,y) = 2 - x - y$$

Since $f_{XY}(x,y) \neq f_X(x) f_Y(y)$, X and Y are **not independent!**

Independence Among RVs - Continuous Case: Example II

Q12. Consider the CRVs X and Y in Question 7 with the joint PDF:

$$f_{XY}(x,y) = \begin{cases} 8xy, & 0 \leq x \leq 1,\ 0 \leq y \leq x \\ 0, & \text{elsewhere} \end{cases}$$

for which we have computed the marginal PDFs as:

$$f_X(x) = 4x^3, \quad x \in [0,1]$$
$$f_Y(y) = 4y - 4y^3, \quad y \in [0,1]$$

Are X and Y independent?

Independence Among RVs - Continuous Case: Example II

Q12. Consider the CRVs X and Y in Question 7 with the joint PDF:

$$f_{XY}(x,y) = \begin{cases} 8xy, & 0 \le x \le 1,\ 0 \le y \le x \\ 0, & \text{elsewhere} \end{cases}$$

for which we have computed the marginal PDFs as:

$$f_X(x) = 4x^3, \quad x \in [0,1]$$
$$f_Y(y) = 4y - 4y^3, \quad y \in [0,1]$$

Are X and Y independent?

A12. Without checking the independence condition, we can simply say that Y can not take any value greater than X. For example, if $X = 0.5$, Y can only get values between 0 and 0.5. **Obviously they are not independent!** But one can also check the independence condition to get the same conclusion.

Independence Among RVs - Continuous Case: Example III

Q13. Consider the following joint PDF of X and Y:

$$f_{XY}(x,y) = 12x^2y^3, \quad 0 < x < 1,\ 0 < y < 1$$

Are X and Y independent?

Independence Among RVs - Continuous Case: Example III

Q13. Consider the following joint PDF of X and Y:

$$f_{XY}(x,y) = 12x^2y^3, \quad 0 < x < 1,\ 0 < y < 1$$

Are X and Y independent?

A13. Let us compute the marginal PDFs of X and Y.

$$f_X(x) = \int_0^1 12x^2y^3 dy = \left[3x^2y^4\right]_{y=0}^{y=1} = 3x^2, \quad x \in [0,1]$$

$$f_Y(y) = \int_0^1 12x^2y^3 dx = \left[4x^3y^3\right]_{x=0}^{x=1} = 4y^3, \quad y \in [0,1]$$

The independence condition holds, thus, we say X and Y are **independent.**

$$f_X(x) f_Y(y) = \left(3x^2\right)\left(4y^3\right) = f_{XY}(x,y) = 12x^2y^3$$

END OF LECTURE SLIDE 08